CPTG: Cosmological Scale Relation

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Abstract

In Curvature Polarization Transport Gravity, the Hubble scale is not treated only as an observational expansion parameter. It is treated as a large-scale structural quantity of the curvature field. Likewise, the internal acceleration scale of a gravitational system is not treated as a universal constant imposed identically on every galaxy. It is a solved structural scale determined by the system's baryonic source distribution, curvature polarization, and transport response.

This paper introduces the CPTG cosmological scale relation as a connection between the local structural acceleration scale and the cosmological structural expansion scale. The central claim is not that every galaxy contains the same acceleration constant, nor that the Hubble scale is numerically fitted from galaxy dynamics. Rather, CPTG proposes that both quantities arise from curvature structure at different scales. The local system expresses curvature structure as an acceleration scale, while the cosmological field expresses curvature structure as an expansion scale.

This interpretation gives CPTG a natural role in the Hubble tension. The disagreement between early-universe and late-universe determinations of the Hubble scale may indicate that the field is being sampled in different curvature-structure regimes, rather than revealing only an observational or calibration conflict. The same structure-dependent mechanism also provides a possible path to connect the expansion-rate tension with the recurring low-growth tendency commonly discussed through the parameter

$$S_8.$$

The resulting scale relation is not a lattice rule or a forced numerical identity. It is a structural relation between local curvature response, cosmological expansion, cosmic-web transport evolution, and the redshift dependence of solved gravitational systems.

1 Introduction

The Hubble scale is usually introduced as a cosmological expansion parameter. It describes the present relationship between distance and recession velocity in the large-scale universe. In standard cosmology, the Hubble constant is inferred through observational programs and interpreted within a background cosmological model.

Galaxy dynamics are usually treated as a separate problem. Rotation curves, dispersion-supported systems, cluster-merger lensing, and large-scale expansion are modeled as distinct regimes. In the standard picture, galaxy and cluster discrepancies are usually attributed to an enlarged matter sector, while the Hubble scale is treated as a background expansion quantity.

Curvature Polarization Transport Gravity proposes a different connection. In CPTG [?], the acceleration scale appearing inside a solved gravitational system is not merely a universal empirical constant. It is a structural scale selected by the system's baryonic source field, curvature polarization, and transport response.

The guiding question of this paper is whether the local acceleration scale and the Hubble scale are separate constants, or whether they are different expressions of curvature structure at different scales.

CPTG takes the second position. The local acceleration scale is interpreted as the bound-system expression of curvature structure. The Hubble scale is interpreted as the cosmological expression of curvature structure. The purpose of the present paper is to formulate this relation in a focused cosmological setting and to identify direct observational tests.

2 Structural Scale Definitions

The local structural acceleration scale is denoted by

$$a_\star. \tag{1}$$

The cosmological structural expansion scale is denoted by

$$H_0. \tag{2}$$

The key CPTG claim is not that these two quantities are the same number. The claim is that they belong to the same curvature-scale family. The local scale describes how a bound system expresses curvature response as acceleration. The cosmological scale describes how the large-scale field expresses curvature structure as expansion.

The dimensional bridge between the two scales is

$$H_\star = \frac{a_\star}{c}. \tag{3}$$

This quantity has units of inverse time,

$$[H_\star] = T^{-1}. \tag{4}$$

The Hubble scale has the same dimensional class,

$$[H_0] = T^{-1}. \tag{5}$$

Dimensional agreement does not imply forced identity. CPTG therefore does not require

$$H_\star = H_0 \tag{6}$$

for every gravitational system.

Instead, the relation is structural. The local scale and the cosmological scale are interpreted as two expressions of curvature organization in different regimes.

3 The Local Structural Acceleration Scale

In CPTG, the local acceleration scale is not imposed identically on every gravitational system. It is treated as a solved scale associated with the structure of the system being evaluated.

A schematic representation is

$$a_\star = \mathcal{A}\left(\mathcal{S}_{\rm local}\right), \tag{7}$$

where

$$\mathcal{S}_{\rm local} \tag{8}$$

represents the local curvature structure of the bound system.

A more explicit structural form is

$$a_\star = \mathcal{A}\left(\mathcal{S}_{\rm local}, \rho_b, R, \mathcal{R}_{PT}\right). \tag{9}$$

Here,

$$\rho_b \tag{10}$$

represents the baryonic source distribution,

$$R \tag{11}$$

represents a characteristic system scale, and

$$\mathcal{R}_{PT} \tag{12}$$

represents the polarization and transport response.

This definition separates CPTG from models that begin with a fixed phenomenological acceleration constant. The acceleration scale is not inserted as a universal interpolation value. It is solved from the source structure and the curvature response of the system.

The local scale therefore belongs to the theory in the same sense as the curvature invariant belongs to the theory. It is not a catalog label, and it is not merely a fitting coefficient. It is the acceleration-scale expression of local curvature organization.

4 Empirical Role of the Local Scale

The local acceleration scale is not introduced only as a dimensional bridge to cosmology. It already plays an active role in the solved structure of bound systems.

Across the current CPTG numerical program, the same scale logic has shown structural consistency across several independent regimes: galaxy-scale rotation systems, merger-scale systems, and the recovered structural mode.

The structural mode is denoted by

$$N. \tag{13}$$

In the galaxy limit, the mode is a curvature-weighted readout of the solved field. It is not a photometric morphology label. It is a dimensionless diagnostic of how the solution organizes curvature support across the system.

A representative structural definition is

$$N = \frac{R}{\lambda}, \tag{14}$$

where

$$\lambda \tag{15}$$

is the characteristic curvature-organization length of the solved field.

The curvature structure can be represented through an invariant of the form

$$C(r) = g^2(r) + \kappa_{\text{struct}}^2 \left(\frac{dg}{dr}\right)^2. \tag{16}$$

The outer weighting field is

$$w(r) = \left|\frac{dC}{dr}\right|. \tag{17}$$

The structural length may then be written as

$$\lambda = \frac{\int_{R/5}^{R} r w(r)\, dr}{\int_{R/5}^{R} w(r)\, dr}. \tag{18}$$

The mode therefore follows from

$$N = \frac{R}{\lambda}. \tag{19}$$

The lower integration limit is chosen as a stable outer-field tracking cutoff:

$$r = \frac{R}{5}. \tag{20}$$

This suppresses core-dominated gradients and isolates the long-range curvature-organization scale used in the galaxy-limit structural mode. For compact systems whose dominant curvature reorganization occurs inside this radius, the cutoff should be re-evaluated or replaced by a system-specific Jeans or effective-radius diagnostic rather than treated as a universal inner-structure measure.

The important point for the present article is that the same local scale concept does not appear only in one galaxy fit. It appears as part of the broader structural behavior of the solved CPTG field.

If the local scale only improved rotation-curve fits, its connection to cosmology would be weak. But if the same scale logic appears across galaxy rotation systems, merger-scale curvature response, and structural mode recovery, then it becomes reasonable to ask whether the cosmological scale is the large-scale counterpart of the same curvature principle.

The conceptual bridge is

$$\text{local curvature structure} \longrightarrow a_\star, \tag{21}$$

and

$$\text{cosmological curvature structure} \longrightarrow H_0. \tag{22}$$

CPTG therefore does not connect the local acceleration scale and the Hubble scale through numerical coincidence. It connects them because both are interpreted as structural scales selected by curvature geometry.

5 The Cosmological Structural Expansion Scale

In standard cosmology, the Hubble constant is usually treated as the present-day expansion rate of the universe. In CPTG, this description is not rejected, but it is incomplete.

CPTG treats the Hubble scale as a structural property of the cosmological curvature field. It is not merely a number assigned to expansion after observations are made. It represents the large-scale rate at which curvature structure expresses itself through cosmic expansion.

The cosmological scale may be written as

$$H_0 = \mathcal{H}\left(\mathcal{S}_{\rm cos}\right), \tag{23}$$

where

$$\mathcal{S}_{\rm cos} \tag{24}$$

represents the large-scale curvature structure of the cosmological field.

This places the Hubble scale in the same conceptual family as the local acceleration scale. The local system expresses curvature structure as acceleration. The cosmological field expresses curvature structure as expansion.

The relation may also be written in redshift-dependent form. Let the cosmological curvature structure at redshift be denoted by

$$\mathcal{S}_{\cos}(z). \tag{25}$$

Then the expansion scale at that epoch is

$$H(z) = \mathcal{H}\left(\mathcal{S}_{\cos}(z)\right). \tag{26}$$

The present-day value is the special case

$$H_0 = H(z = 0). \tag{27}$$

Thus, in CPTG, the Hubble scale is not only a background number. It is the present expression of a curvature-structure history.

6 The CPTG Cosmological Scale Relation

The CPTG cosmological scale relation proposes that local gravitational acceleration scales and the cosmological Hubble scale are connected because both arise from curvature structure.

For a bound system, the local solved acceleration scale is

$$a_\star = \mathcal{A}\left(\mathcal{S}_{\text{local}}\right). \tag{28}$$

For the cosmological field, the large-scale expansion structure is

$$H_0 = \mathcal{H}\left(\mathcal{S}_{\cos}\right). \tag{29}$$

The local inverse-time expression is

$$H_\star = \frac{a_\star}{c}. \tag{30}$$

CPTG does not require the identity

$$H_\star = H_0 \tag{31}$$

for every system.

Such a requirement would erase the structural differences among galaxies, clusters, compact spheroids, slow rotators, and large-scale environments. It would reduce the theory back to a universal acceleration-law model.

Instead, CPTG proposes that

$$H_\star \tag{32}$$

and

$$H_0 \tag{33}$$

belong to the same curvature-scale family.

The first is the local time-scale expression of curvature response inside a bound system. The second is the global time-scale expression of curvature structure across the cosmological field.

The theory's claim is therefore not that one number is trivially substituted for the other. The claim is that both scales are generated by the same underlying curvature-transport principle.

7 Hubble Tension as a Structural-Scale Problem

The Hubble tension is usually presented as a disagreement between early-universe and late-universe determinations of the present expansion rate. Planck base Lambda-CDM inference gives a lower value of the Hubble scale, while the SH0ES distance-ladder program reports a higher local value.

Representative values are

$$H_0^{\text{Planck}} = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}, \tag{34}$$

and

$$H_0^{\text{SH0ES}} = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}. \tag{35}$$

In CPTG, the Hubble tension is interpreted as a possible structural-scale mismatch. The early-universe value is not merely a direct measurement of the present local expansion rate. It is inferred by evolving early curvature information through a cosmological model. The late-universe value is measured from local distance indicators and supernova calibration.

CPTG treats these as probes of different curvature regimes.

The early-universe inference may be written schematically as

$$H_{\text{early}} = \mathcal{H}\left(\mathcal{S}_{\text{early}}\right). \tag{36}$$

The late-universe measurement may be written as

$$H_{\text{late}} = \mathcal{H}\left(\mathcal{S}_{\text{late}}\right). \tag{37}$$

The tension appears when both are forced into a single structural value without accounting for possible curvature-transport evolution between regimes.

The standard identification can be represented as

$$\mathcal{H}\left(\mathcal{S}_{\text{early}}\right) \longrightarrow H_0, \tag{38}$$

and

$$\mathcal{H}(\mathcal{S}_{\text{late}}) \longrightarrow H_0. \tag{39}$$

CPTG allows the possibility that the early and late values are related but not structurally identical. The disagreement may therefore indicate that the curvature field has undergone transport evolution not fully represented by the standard expansion model.

This does not mean CPTG can simply declare the Hubble tension solved. The theory must show that its structural relation can connect early expansion, late expansion, galaxy-scale acceleration structure, and redshift evolution without arbitrary fitting.

The Hubble tension nevertheless becomes a direct motivation for CPTG. It suggests that the universe may not be fully described by a single expansion parameter alone, but by a curvature structure whose scale evolves across cosmic time.

8 Redshift Evolution of the Local Scale

If the local acceleration scale and the cosmological expansion scale are both structural curvature quantities, then the local scale should not be assumed to be completely fixed across cosmic history.

The simplest CPTG redshift scaling is

$$a_\star(z) = a_\star(0)\frac{H(z)}{H_0}. \tag{40}$$

This expression does not mean that every galaxy at the same redshift has the same internal acceleration scale. Rather, it means that the background curvature scale participates in the structural environment from which each system solves its own local response.

A more complete expression preserves both local and cosmological dependence:

$$a_\star(z) = \mathcal{A}\left(\mathcal{S}_{\text{local}}(z), \mathcal{S}_{\cos}(z)\right). \tag{41}$$

The redshift scaling provides a direct testable bridge. If the background expansion scale is larger at high redshift, then the structural acceleration scale available to bound systems should also be larger, after controlling for baryonic structure and system size.

This leads directly to the high-redshift slow-rotator test developed below.

9 Growth Coupling and the Low-Growth Tendency

The Hubble tension is not the only large-scale anomaly motivating a structural interpretation of cosmology. Late-time structure measurements have also shown a recurring tendency toward lower matter-clustering amplitude than the value inferred from the cosmic microwave background under base Lambda-CDM assumptions.

The structure-growth parameter is commonly written as

$$S_8 = \sigma_8 \sqrt{\frac{\Omega_m}{0.3}}. \tag{42}$$

Planck base Lambda-CDM gives a matter fluctuation amplitude near

$$\sigma_8 = 0.811 \pm 0.006. \tag{43}$$

Weak-lensing and galaxy-clustering surveys often favor lower clustering amplitudes. This tendency is survey-dependent and should not be overstated as a settled crisis. In the present paper it is treated as a growth-amplitude tendency rather than as a definitive failure of the standard model.

CPTG suggests that the expansion tension and the growth-amplitude tendency may not be separate problems. If the Hubble scale is a structural curvature scale, then the growth of matter structure is also affected by the same transport geometry.

Because the local curvature scale evolves as

$$a_\star(z) = a_\star(0)\frac{H(z)}{H_0}, \tag{44}$$

the threshold for curvature-polarization response is not fixed across cosmic time. At higher redshift, the larger background expansion scale raises the effective structural acceleration scale available during matter assembly.

This shifts the epoch at which matter perturbations enter the polarization-dominated or transport-modified regime. The late-time growth amplitude is therefore not independent of the same curvature-scale evolution that enters the Hubble relation.

Let the cosmological scale factor be

$$a_{\cos} = \frac{1}{1+z}. \tag{45}$$

A standard-form schematic growth equation may be written as

$$\frac{d^2\delta}{d\ln^2 a_{\cos}} + \left(2 + \frac{d\ln H}{d\ln a_{\cos}}\right)\frac{d\delta}{d\ln a_{\cos}} = \mathcal{G}_{\rm CPTG}(z)\delta. \tag{46}$$

The CPTG growth source can be expressed as

$$\mathcal{G}_{\rm CPTG}(z) = \frac{3}{2}\Omega_m(z)\mu_{\rm CPTG}(z). \tag{47}$$

The effective response factor is written as

$$\mu_{\rm CPTG}(z) = \mathcal{R}_{PT}(z) - \mathcal{D}_{\rm trans}(z). \tag{48}$$

Here,

$$\mathcal{R}_{PT}(z) \tag{49}$$

represents the curvature-polarization response, while

$$\mathcal{D}_{\rm trans}(z) \tag{50}$$

represents transport redistribution, environmental leakage, or screening of clustered growth.

A minimal polarization-response closure can be written as

$$\mathcal{R}_{PT}(z) = \left[1 + \left(\frac{a_\star(z)}{g_{\cos}(z)}\right)^2\right]^{1/4}. \tag{51}$$

Here,

$$g_{\cos}(z) \tag{52}$$

is a characteristic cosmic-web acceleration scale. It is not intended as an unconstrained free parameter. In the simplest implementation, it should be tied to a background or smoothed cosmic-web field derived from the matter distribution and expansion history, with baseline density scaling controlled by

$$\rho_m(z) = \rho_{m,0}(1+z)^3. \tag{53}$$

The precise operational definition of

$$g_{\cos}(z) \tag{54}$$

should be fixed before numerical growth tests are interpreted.

For the first numerical implementation, a controlled power-law closure may be used:

$$g_{\cos}(z) = g_{\cos,0}(1+z)^{\nu}. \tag{55}$$

A filament-like geometric scaling suggests

$$\nu \simeq 2, \tag{56}$$

while a density-square-root scaling suggests

$$\nu \simeq \frac{3}{2}. \tag{57}$$

The exponent should be treated as a fixed test choice or a tightly bounded closure parameter, not as an unconstrained fit.

This formulation is intentionally conservative. Curvature polarization alone can strengthen the effective response. A lower late-time value of the growth amplitude requires that transport redistribution or screening suppress the amount of matter remaining in clustered form. Thus CPTG does not assume the desired value of

$$S_8. \tag{58}$$

It supplies a calculable mechanism by which the same redshift-dependent structural scale that enters the Hubble relation can also modify the growth history.

A successful calculation must reproduce the observed late-time growth amplitude while using the same curvature-scale evolution that enters the Hubble-scale relation.

10 Cosmic-Web Transport Evolution

CPTG interprets the difference between early and late cosmological scales as a consequence of transport evolution in the curvature field.

The early universe was close to homogeneous and isotropic. Its curvature structure was nearly uniform, with small perturbations that later seeded galaxies, clusters, filaments, sheets, and voids. The late universe is different. It contains a highly developed cosmic web, with large density contrasts and strong environmental separation between collapsed structures, filaments, sheets, and voids.

This transition changes the structure sampled by the cosmological field.

The early structural state may be written as

$$\mathcal{S}_{\text{early}} \simeq \mathcal{S}_{\text{uniform}} + \delta\mathcal{S}. \tag{59}$$

The late structural state is better represented as

$$\mathcal{S}_{\text{late}} = \mathcal{S}_{\text{void}} + \mathcal{S}_{\text{filament}} + \mathcal{S}_{\text{sheet}} + \mathcal{S}_{\text{halo}}. \tag{60}$$

The emergence of strong spatial curvature gradients can be represented schematically by

$$\partial_i g_j \neq 0. \tag{61}$$

The notation

$$\partial_i g_j \tag{62}$$

is used here as a local quasi-static spatial projection of the corresponding covariant curvature-gradient structure. In a fully covariant treatment, the spatial expression should be read schematically as the projected component of a covariant derivative structure such as

$$\Pi_i{}^{\alpha}\Pi_j{}^{\beta}\nabla_{\alpha}G_{\beta}. \tag{63}$$

Thus the present section is an effective cosmological reduction, not a complete covariant closure of the transport sector.

In CPTG, these gradients do not merely trace matter structure after the fact. They participate in the transport response of the curvature field.

The transport vector may be written as

$$S^{\alpha} = S^{\alpha}\left(\mathcal{S}_{\text{cos}}, \partial_i g_j, \rho_b, z\right). \tag{64}$$

As the cosmic web forms, the transport field changes regime. The present paper describes this as a transport-regime transition rather than as a literal thermodynamic phase transition.

A minimal directional closure for the transport field is

$$S^{\alpha} = \Xi_{\text{web}}(z)\frac{g^{\alpha\beta}\partial_{\beta}\mathcal{C}_{\text{cos}}}{|\nabla\mathcal{C}_{\text{cos}}|}. \tag{65}$$

Here,

$$\mathcal{C}_{\text{cos}} \tag{66}$$

is the large-scale curvature invariant, and

$$\Xi_{\mathrm{web}}(z) \tag{67}$$

is a web-activation kernel.

A simple transition kernel is

$$\Xi_{\mathrm{web}}(z) = \frac{1}{1 + \exp\left(\frac{z - z_{\mathrm{web}}}{\Delta z}\right)}. \tag{68}$$

At early times,

$$z > z_{\mathrm{web}}, \tag{69}$$

the universe is closer to homogeneous and the directional transport channel is weak.

At late times,

$$z < z_{\mathrm{web}}, \tag{70}$$

the cosmic web has developed strong void, filament, sheet, and halo structure, and the transport channel becomes active.

The CPTG interpretation is

$$\mathcal{S}_{\mathrm{early}} \longrightarrow \mathcal{S}_{\mathrm{late}} \tag{71}$$

through cosmic-web formation.

This provides a physical basis for why early-universe and late-universe probes may not return the same inferred structural Hubble scale. They are not sampling the same curvature-transport regime.

11 Environmental Screening and the Distance Ladder

Late-universe measurements of the Hubble scale rely on distance indicators embedded in real astrophysical environments. Cepheids, supernovae, host galaxies, calibrator systems, and the larger-scale environments around those systems are not isolated from curvature structure.

CPTG therefore introduces the possibility of environmental screening. The local acceleration scale was written above as

$$a_\star = \mathcal{A}(\mathcal{S}_{\mathrm{local}}, \rho_b, R, \mathcal{R}_{\mathrm{PT}}). \tag{72}$$

For cosmological distance-ladder systems, the local structure should be extended to include the background cosmological environment:

$$a_\star = \mathcal{A}(\mathcal{S}_{\mathrm{local}}, \mathcal{S}_{\mathrm{cos}}, \rho_b, R, \mathcal{R}_{\mathrm{PT}}). \tag{73}$$

The effective local scale is written as

$$a_{\star,\mathrm{eff}} = \mathcal{E}_{\mathrm{env}} a_{\star}. \tag{74}$$

The environmental operator satisfies

$$0 < \mathcal{E}_{\mathrm{env}} \leq 1. \tag{75}$$

Let

$$Q_g = (\partial_i g_j)(\partial_i g_j). \tag{76}$$

A minimal bounded screening operator is

$$\mathcal{E}_{\mathrm{env}} = \frac{1}{1 + \left(\frac{\rho_{\mathrm{env}}}{\rho_{\mathrm{crit}}}\right)^p \left(\frac{\kappa_{\mathrm{struct}}^2 Q_g}{a_\star^2}\right)^q}. \tag{77}$$

Here, ρ_{env} is the large-scale environmental density, and Q_g measures local curvature-gradient structure.

In low-density, weak-gradient environments, the screening operator approaches

$$\mathcal{E}_{\mathrm{env}} \to 1. \tag{78}$$

The local structural scale remains active. In dense, high-gradient environments, the screening operator decreases:

$$\mathcal{E}_{\mathrm{env}} < 1. \tag{79}$$

The local polarization response is partially screened.

The observational prediction is direct:

$$\Delta H_{0,\mathrm{ladder}} = \Delta H_0(\rho_{\mathrm{env}}, Q_g, \mathcal{E}_{\mathrm{env}}). \tag{80}$$

If CPTG is correct, then residuals in local distance-ladder measurements should correlate with environmental density and curvature-gradient structure after ordinary calibration effects are controlled.

12 High-Redshift Slow Rotators as a Cross-Regime Test

A direct test of the CPTG cosmological scale relation is provided by massive high-redshift quiescent galaxies and slow rotators.

These systems are valuable because they are not ordinary rotation-curve galaxies. Their gravitational fields are inferred through pressure support, velocity dispersion, stellar density, and effective radius. They therefore test whether the CPTG local structural scale applies beyond disk rotation systems.

The high-redshift slow rotator XMM-VID1-2075 provides an example of this kind of test. In the effective-radius Jeans validation, the local scale was solved from the dispersion-supported structure rather than from rotation.

The redshift is approximately

$$z \simeq 3.449. \tag{81}$$

The observed dispersion input is

$$\sigma_\star = 387 \pm 22 \text{ km s}^{-1}. \tag{82}$$

The effective radius used in the reduced validation is

$$R_e = 2.00 \text{ kpc}. \tag{83}$$

The effective-radius acceleration scale is

$$a_\star = \frac{\sigma_\star^2}{R_e}. \tag{84}$$

Using the central observed dispersion gives

$$a_\star = 2.42684 \times 10^{-9} \text{ m s}^{-2}. \tag{85}$$

The baryonic spheroid field at the effective radius is

$$g_\star(R_e) = 4.78107 \times 10^{-9} \text{ m s}^{-2}. \tag{86}$$

The reduced polarization and transport response at the effective radius is

$$\mathcal{R}_{PT}(R_e) = 1.40378. \tag{87}$$

The corresponding CPTG effective-radius field is

$$g_{\text{CPTG}}(R_e) = 6.71157 \times 10^{-9} \text{ m s}^{-2}. \tag{88}$$

The recovered structural mode at the effective radius is

$$N_{R_e} = 2.76556. \tag{89}$$

The finite-zone corrected dispersion is

$$\sigma_{\text{CPTG,finite}} = 395.16 \text{ km s}^{-1}. \tag{90}$$

The core corrected dispersion is

$$\sigma_{\text{CPTG,core}} = 393.76 \text{ km s}^{-1}. \tag{91}$$

Both dispersion estimates fall inside the observed interval.

For a pressure-supported system, the effective-radius scale may be written schematically as

$$a_\star \sim \frac{\sigma_\star^2}{R_e}. \tag{92}$$

At high redshift, CPTG predicts that the structural acceleration scale should be evaluated in the cosmological environment of that epoch:

$$a_\star(z) = a_\star(0)\frac{H(z)}{H_0}. \tag{93}$$

Because the expansion scale satisfies

$$H(z) > H_0 \tag{94}$$

at high redshift, the available curvature-response scale should be amplified relative to the present epoch.

The observational prediction is

$$\sigma_\star^2 \sim a_\star(z) R_e. \tag{95}$$

Equivalently,

$$\sigma_\star \sim \sqrt{a_\star(z) R_e}. \tag{96}$$

This gives CPTG a clean cross-regime test. Massive quiescent galaxies at high redshift should show velocity-dispersion structure consistent with the amplified curvature scale of their epoch.

12.1 Fixed-Structure Redshift Scaling

The redshift prediction becomes quantitative when the high-redshift object is compared to a same-structure present-epoch analog.

At fixed effective radius and fixed baryonic structure,

$$\frac{\sigma_\star(z)}{\sigma_\star(0)} = \left(\frac{H(z)}{H_0}\right)^{1/2}. \tag{97}$$

For a flat matter-plus-Lambda background,

$$\frac{H(z)}{H_0} = \left(\Omega_m(1+z)^3 + \Omega_\Lambda\right)^{1/2}. \tag{98}$$

Therefore,

$$\frac{\sigma_\star(z)}{\sigma_\star(0)} = \left(\Omega_m(1+z)^3 + \Omega_\Lambda\right)^{1/4}. \tag{99}$$

For XMM-VID1-2075, using

$$z = 3.449, \tag{100}$$

$$\Omega_m = 0.315, \tag{101}$$

and

$$\Omega_\Lambda = 0.685, \tag{102}$$

one obtains

$$\frac{H(z)}{H_0} \simeq 5.33. \tag{103}$$

The corresponding fixed-structure dispersion amplification is

$$\frac{\sigma_\star(z)}{\sigma_\star(0)} \simeq 2.31. \tag{104}$$

Thus the observed high-redshift dispersion

$$\sigma_\star(z) = 387 \text{ km s}^{-1} \tag{105}$$

corresponds to a same-structure present-epoch analog of approximately

$$\sigma_\star(0) \simeq 168 \text{ km s}^{-1}. \tag{106}$$

The measured high-redshift acceleration scale

$$a_\star(z) = 2.42684 \times 10^{-9} \text{ m s}^{-2} \tag{107}$$

corresponds to a same-structure present-epoch analog of

$$a_\star(0) \simeq 4.55 \times 10^{-10} \text{ m s}^{-2}. \tag{108}$$

These values should not be interpreted as a claim that XMM-VID1-2075 is literally the same object as a local galaxy with lower dispersion. The calculation is a fixed-structure redshift scaling. Real high-redshift systems must also be corrected for compactness, stellar population, baryonic mass, effective radius, and selection effects.

The prediction is nevertheless sharp. At fixed source structure, CPTG predicts

$$\sigma_\star(z) \propto H^{1/2}(z). \tag{109}$$

A future sample of high-redshift quiescent galaxies and slow rotators can therefore test whether dispersion-supported systems track the amplified curvature scale of their epoch.

13 Numerical Test Program

The cosmological scale relation is meaningful only if it can be tested without allowing every regime to float independently. The present formulation suggests four linked numerical tests.

First, the redshift scaling test should solve

$$a_\star(z) = a_\star(0)\frac{H(z)}{H_0} \tag{110}$$

across a sample of high-redshift dispersion-supported systems. The target observable is

$$\sigma_\star(z) \tag{111}$$

at fixed or controlled effective radius, mass, and baryonic structure.

Second, the growth test should solve

$$\frac{d^2\delta}{d\ln^2 a_{\cos}} + \left(2 + \frac{d\ln H}{d\ln a_{\cos}}\right)\frac{d\delta}{d\ln a_{\cos}} = \mathcal{G}_{\mathrm{CPTG}}(z)\delta \tag{112}$$

using the same redshift-dependent structural scale. The target observable is the late-time growth amplitude.

Third, the environmental screening test should regress local distance-ladder residuals against

$$\rho_{\mathrm{env}} \tag{113}$$

and

$$Q_g. \tag{114}$$

The target observable is

$$\Delta H_{0,\mathrm{ladder}}. \tag{115}$$

Fourth, the cosmic-web transition test should fit the web-activation kernel

$$\Xi_{\mathrm{web}}(z) \tag{116}$$

against large-scale structure data. The target is the redshift interval over which curvature transport becomes dynamically organized.

The important requirement is that these tests must not be tuned independently. The same structural scale evolution must support the high-redshift dispersion relation, the growth history, and the environmental residual pattern.

14 Scope and Falsifiability

The CPTG cosmological scale relation is not a replacement for direct observational measurement of the Hubble parameter. It is a proposed structural relation between local gravitational response and global expansion geometry.

The relation can be challenged if independently solved systems show no stable relationship between local acceleration structure and cosmological curvature scale.

It can also be challenged if redshift evolution fails to show the expected structural dependence.

A valid test must distinguish between

$$\text{physical curvature structure,} \tag{117}$$

$$\text{numerical construction,} \tag{118}$$

and

$$\text{statistical coincidence.} \tag{119}$$

The theory gains strength only if the same structural relation appears across independent systems, independent datasets, and independent solution methods.

The strongest falsification path is therefore not a single galaxy or a single fitted value. It is a cross-regime test. CPTG should be evaluated across galaxy rotation systems, dispersion-supported systems, merger-scale systems, structural mode recovery, redshift-dependent slow rotators, environmental distance-ladder residuals, and large-scale growth measurements.

If the local scale continues to appear as a solved structural quantity across these regimes, and if the cosmological scale can be interpreted as the large-scale expression of the same curvature principle, then the scale relation becomes physically meaningful.

If those cross-regime connections fail, then the cosmological scale relation should be rejected or substantially revised.

15 Conclusion

The CPTG cosmological scale relation proposes that the local acceleration scale and the Hubble scale are not unrelated empirical constants. They are structural curvature scales expressed in different regimes.

The local acceleration scale is

$$a_\star. \tag{120}$$

It is the bound-system expression of solved curvature structure.

The Hubble scale is

$$H_0. \tag{121}$$

It is the cosmological expression of large-scale curvature structure.

The dimensional bridge is

$$\frac{a_\star}{c}. \tag{122}$$

This bridge shows that the local acceleration scale can be expressed as an inverse-time quantity, placing it in the same dimensional family as the Hubble scale. But CPTG does not require a forced universal equality between them.

Instead, CPTG proposes a shared geometric origin. Local systems express curvature structure through acceleration. The cosmological field expresses curvature structure through expansion.

This gives CPTG a natural role in the Hubble tension. The early and late values of the Hubble scale may differ because they probe different stages or regimes of curvature-transport structure. The tension may therefore be not only a measurement problem, but a structural clue.

The same logic extends to structure growth. If the expansion scale and the matter-growth amplitude are both shaped by transport evolution of the curvature field, then the Hubble tension and the low-growth tendency should not be treated as fully independent anomalies.

The strongest immediate quantitative test is the redshift scaling of dispersion-supported systems. For high-redshift slow rotators, CPTG predicts

$$\sigma_\star(z) \propto H^{1/2}(z) \tag{123}$$

at fixed source structure. XMM-VID1-2075 provides an initial effective-radius Jeans validation of this cross-regime direction, and future high-redshift quiescent-galaxy samples can test whether the same scaling persists.

In this interpretation, CPTG does not merely add a new parameter to cosmology. It proposes that galaxy dynamics, merger-scale curvature response, structural mode recovery, cosmic expansion, late-time growth, and high-redshift dispersion structure are connected by the same underlying curvature-scale principle.

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